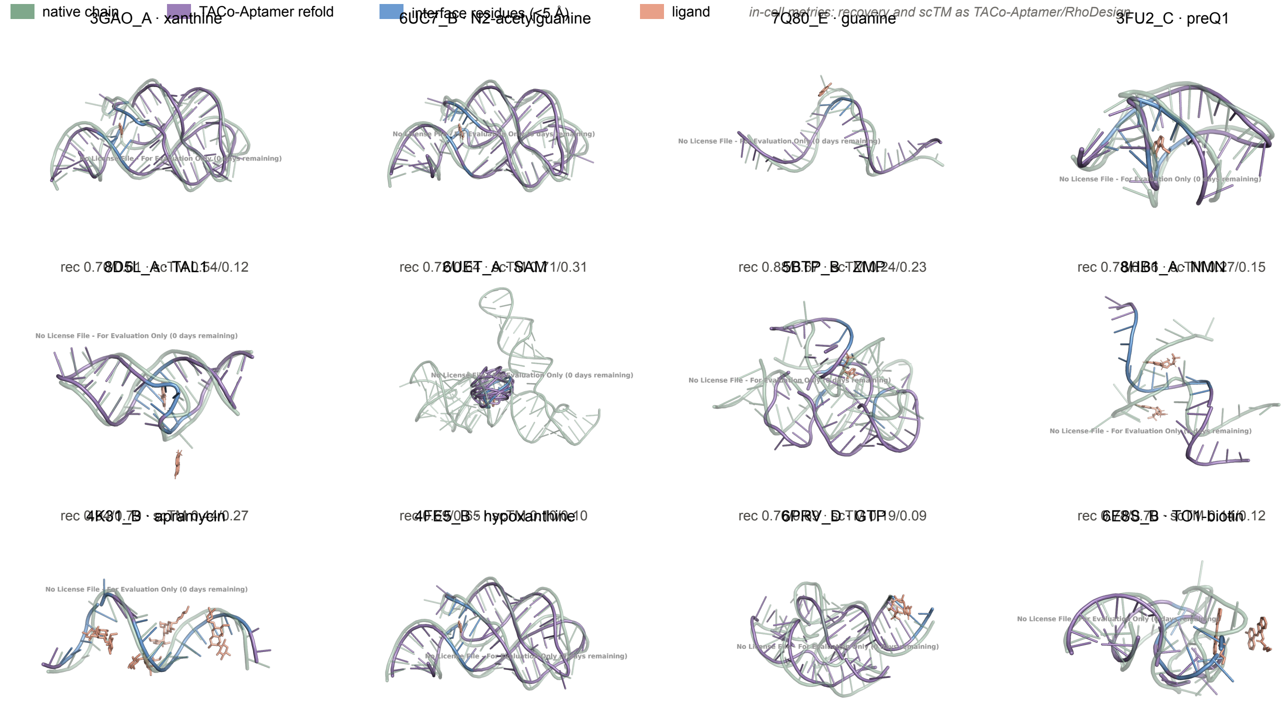
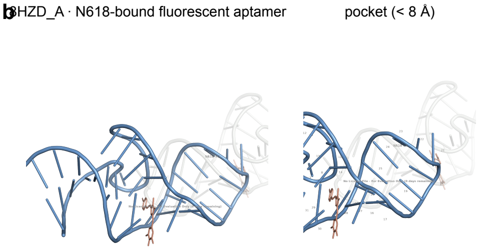
**a**

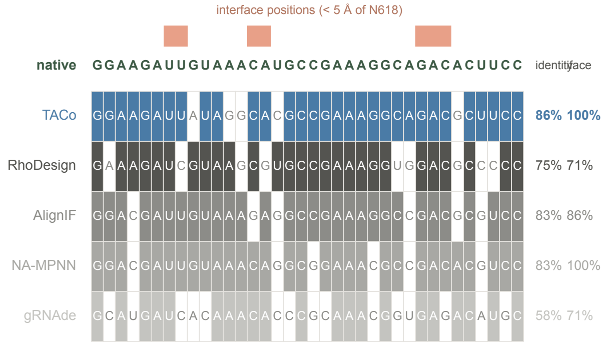

native chain
TACo-Aptamer refold
interface residues (<5 Å)
ligand
in-cell metrics: recovery and scTM as TACo-Aptamer/RhoDesign
3GAO_A · xanthine
6UC7_B · N2-acetylguanine
7Q80_E · guanine
3FU2_C · preQ1
8D5L_A · TAL1
8UE4_A · SAM
5BTP_B · ZMP
8HB1_A · NMN
4K31_B · apramycin
4FE5_B · hypoxanthine
6PRV_D · GTP
6E8S_B · TOM-biotin
No License File - For Evaluation Only (0 days remaining)
rec 1.00/0.95 · scTM 0.22/0.21
rec 0.70/0.63 · scTM 0.58/0.40
rec 0.72/0.74 · scTM 0.40/0.10
rec 0.68/0.74 · scTM 0.28/0.33

**b**



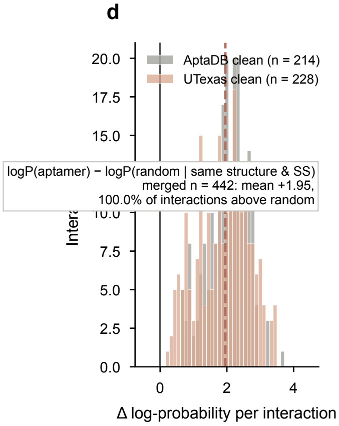
**c**

Best-of-pool designs, 8HZD_A (filled = match to native)


interface positions (< 5 Å of N618)

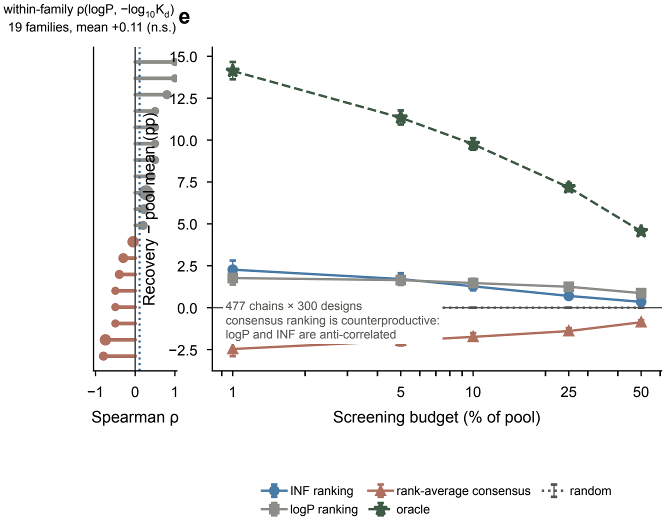
| | sequence | identity | iface |
|---|---|---|---|
| native | GGAAGAUUGUAAACAUGCCGAAAGGCAGACACUUCC | | |
| TACo | GGAAGAUUAUAGGCACGCCGAAAGGCAGACGCUUCC | 86% | 100% |
| RhoDesign | GAAAGAUCGUAAGCGUGCCGAAAGGUGGACGCCCCC | 75% | 71% |
| AlignIF | GGACGAUUGUAAAGAGGCCGAAAGGCCGACGCGUCC | 83% | 86% |
| NA-MPNN | GGACGAUUGUAAACAGGCGGAAACGCCGACACGUCC | 83% | 100% |
| gRNAde | GCAUGAUCACAAACACCCGCAAACGGUGAGACAUGC | 58% | 71% |

**d**



logP(aptamer) − logP(random | same structure & SS)
merged n = 442: mean +1.95,
100.0% of interactions above random

**e**



477 chains × 300 designs
consensus ranking is counterproductive:
logP and INF are anti-correlated